DOD-C-24689A
6 November 1987
SUPERSEDING
DOD-C-24689
11 August 1987
(See 6.4)

MILITARY SPECIFICATION

COBALT ALLOY CASTINGS, WEAR AND CORROSION RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers castings of cobalt-chromium-tungsten alloys for use where corrosion and wear resistant materials are required.

1.2 Classification. Castings shall be of the following compositions, as specified (see 6.2.1).

Composition No. 20 (UNS No. R30006)
Composition No. 21
Composition No. 22 (UNS No. R30001)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY
MIL-C-3993 - Copper and Copper-Base Alloy Mill Products; Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-151 - Metals; Test Methods.

MILITARY

MIL-STD-271 - Requirements for Nondestructive Testing Methods.
MIL-STD-278 - Welding and Casting Standard.
MIL-STD-792 - Identification Marking Requirements for Special Purpose Components.

(Copies of specifications and standards required by contractors in connection with specific aquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 18 - Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials. (DoD adopted)
E 354 - Standard Methods for Chemical Analysis of High Temperature, Electrical, Magnetic, and Other Similar Iron, Nickel, and Cobalt Alloys.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or order, a sample casting shall be subjected to first article inspection (see 4.3 and 6.3).

3.1.1 Basic foundry practices. The contractor shall maintain a record of the basic foundry practices (type of melting and refining unit, molding process, basic rigging design, location of risers and chills) used in the first article casting and shall make no change thereto without the specific

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approval of the command or agency concerned. The contractor may be required to perform specific first article tests and examinations to verify that the change will not degrade casting quality.

3.2 Material. Material shall be made by an electric furnace process. Unless a specific type is specified (see 6.2.1), static, centrifugal or investment castings may be supplied at the option of the producer.

3.3 Process prohibitions. Unspecified heat treatments, cold working, straightening or weld repair shall not be permitted.

3.4 Descaling. Castings shall be cleaned so that the surface is free from scale and foreign matter. If shot, grit or vapor blasting is used, it shall be followed by machining, grinding or pickling (provided a minimum of 0.010 inch (0.25 mm) is removed in subsequent machining) when liquid penetrant inspection is required.

3.5 Recovered materials. Unless otherwise specified herein, all material, used in the products covered by this specification shall be new and may be processed using recovered materials to the maximum extent practicable without jeopardizng the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used products is allowed under this specification unless otherwise specifically specified.

3.6 Chemical composition. The casting material shall conform to the chemical composition specified in table I.

TABLE I. Chemical composition.

Element	Comp 20 (percent)	Comp 21 (percent)	Comp 22 (percent)
Carbon	0.90-1.4	1.8-2.2	2.0-2.6
Manganese	1.0 maximum	0.50 maximum	1.0 maximum
Silicon	1.5 maximum	0.80-1.5	1.5 maximum
Chromium	27.0-31.0	23.0-25.0	29.0-33.0
Tungsten	3.5-5.5	11.0-13.0	11.0-14.0
Nickel	3.0 maximum	3.0 maximum	3.0 maximum
Iron	3.0 maximum	2.0 maximum	3.0 maximum
Molybdenum	1.5 maximum	1.0 maximum	1.0 maximum
Others, total (Ti, As, P, S, Cu, Mo, B and Zn)	1.0 maximum	1.0 maximum	1.0 maximum
Cobalt	Remainder	Remainder	Remainder

3.7 Mechanical properties. The material shall conform to the mechanical properties specified in table II.

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TABLE II. Hardness.

Alloy	Rockwell "C" minimum	
	Static & centrifugal castings	Investment castings
Composition 20	40	37
Composition 21	50	48
Composition 22	50	48

3.8 Heat treatment. Castings shall be stress relieved at 1650 ± 25 degrees Fahrenheit (°F) (900 ± 14 degrees Celsius (°C)) for 1 hour per inch of section with a 2 hour minimum total time at temperature. Castings shall be cooled in the furnace at 250 to 500°F per hour maximum (120 to 260°C per hour maximum), after which the castings may be cooled in still air to ambient temperature. Cooling may be done in steps provided the specified cooling rate is maintained.

3.9 Soundness. Castings shall be of uniform quality and condition, free of cracks, burrs, sand, tears, shrinkage, cold shuts, excessive scale, fins, porosity, and segregation.

3.10 Removal of defects. Defects may be ground out provided the minimum dimensions of the casting drawing are met.

3.11 Dimensions. Castings shall be such that they can be laid out and machined to the finished dimensions, with the specified tolerances, in accordance with the casting drawing. Sufficient stock shall be allowed for shrinkage and, where required, for finishing. Rough castings shall not exceed 1/16 inch (1.5 mm) over specified size and machined castings shall not exceed 3/32 inch (2.4 mm) over specified size.

3.12 Identification. Identification shall be as specified in 3.12.1 through 3.12.3.

3.12.1 Information required. Individual castings shall be marked with the following:

(a) Foundry's name or trademark
(b) Specification number
(c) Pattern or part number
(d) Heat number and heat treat lot number

3.12.2 Methods and location of marking. The methods of marking shall be by electrochemical etching by casting in, by low-stress die stamping, or vibro-etching or in accordance with MIL-STD-792. The location of marking shall be at a low stress area as shown on the casting drawing.

3.12.3 Small castings. Where individual castings cannot be marked as specified in 3.12.1 and 3.12.2, they may be bundled or boxed and each bundle or box shall be marked with a metal or tag containing the information specified in 3.12.1.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificate of compliance. When specified in the contract or order, a certificate of compliance shall be prepared (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the test as specified in 4.3.2.

4.3.1 Sampling for first article inspection. The first castings submitted for inspection shall be the first article sample.

4.3.2 Radiographic inspection. The first article casting shall be radiographically inspected in accordance with MIL-STD-271 to the acceptance criteria specified in MIL-STD-278 of the criticality level specified, or to another standard, as specified (see 6.2.1).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with 4.4.1 through 4.4.2.4.

4.4.1 Lot definition. Lots shall be as defined in 4.4.1.1 through 4.4.1.2.

4.4.1.1 Lot for chemical analysis. Lot for chemical analysis shall be as follows:

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4.4.1.1.1 Master melt used. When a master melt is used, a lot shall consist of all material cast from the same master melt, by the same process, and presented for inspection at the same time.

4.4.1.1.2 Master melt not used. When a master melt is not used, a lot shall consist of all the castings cast from the same heat, by the same process, and presented for inspection at the same time.

4.4.1.1.3 Carbon additions. When carbon additions are made during the pouring of castings for the purpose of carbon analysis only, a lot shall be further defined as the castings poured between carbon additions.

4.4.1.2 Visual and dimensional examination, and mechanical property test. A lot shall consist of all the castings from a lot defined in 4.4.1.1.1 or 4.4.1.1.2 that have the same maximum cross-section and which have been heat treated togther.

4.4.2 Sampling. Sampling for quality conformance shall be as specified in 4.4.2.1 through 4.4.2.4.

4.4.2.1 Sampling for chemical analysis. Two samples shall be selected at random from each lot. Samples may be taken from the ladle, or from casting blanks. When taken from the ladle, the samples shall be poured at the same time the castings are poured. When carbon additions are made to the melt during the pouring of castings, samples shall be taken for carbon analysis as follows: one immediately before each addition and one after each addition, following an appropriate hold time for homogenization.

4.4.2.2 Sampling for hardness tests. Two castings shall be tested from each lot.

4.4.2.3 Sampling for nondestructive inspection. Each casting shall be inspected.

4.4.2.4 Visual and dimensional examination. Each casting shall be examined visually and dimensionally for conformance to this specification.

4.5 Tests. Tests shall be performed as specified in 4.5.1 through 4.5.4.

4.5.1 Chemical analysis. Chemical analysis shall be performed by wet or spectrographic methods as determined by the contractor. In case of dispute, the wet method as specified in ASTM E 354 shall govern.

4.5.2 Hardness test. Hardness tests shall be performed in accordance with ASTM E 18.

4.5.3 Nondestructive testing. When specified on the casting drawing, castings shall be liquid penetrant inspected or radiographed to the extent specified. Testing shall be in accordance with MIL-STD-271.

4.5.4 Visual and dimensional examination. Any casting which fails to meet the dimensional requirement of the casting drawing shall be rejected. Castings shall be examined for conformance to 3.4 and 3.9.

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4.6 Replacement, retest, rejection, and resubmittal. Replacement, retest, rejection, and resubmittal shall be as specified in 4.6.1 through 4.6.5.

4.6.1 Replacement of test specimens. A test specimen (for chemical analysis) or location (for hardness) may be discarded and a replacement test specimen selected from the same lot of material in accordance with FED-STD-151.

4.6.2 Retest. Retesting shall be permitted for chemical analysis and hardness only, as follows:

(a) When one or more specified charateristic fails to conform to the requirements for the tested characteristic, retest of the nonconforming characteristic may be performed to establish product acceptability.
(b) Retesting shall be performed on twice the number of representative specimens that were originally nonconforming.
(c) Unless otherwise approved by the acquisition activity, retest specimens shall be taken adjacent to the initial nonconforming test specimen location.

4.6.3 Rejection. When any retest specimen does not conform to the requirements for the characteristic being tested, the lot represented by the specimen shall be rejected.

4.6.4 Resubmittal of rejected lots. A rejected lot may be resubmitted for acceptance testing only after rework, as necessary, is performed to correct the nonconforming condition without adversely affecting the conforming properties. If the rejected lot is re-heat treated to correct a nonconforming charateristic, all mechanical properties, including those which were initially conforming, shall be determined.

4.6.5 Resubmittal of individual pieces. At the option of the contractor, when a rejected lot consists of more than one piece, each remaining piece in the lot may be tested for the nonconforming characteristic and each piece that conforms to all the requirements specified herein may be offered for acceptance.

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Packaging, packing and marking. The material shall be packaged level A or C; packed level A, B, or C as specified (see 6.2.1) and marked in accordance with MIL-C-3993. Other standards, as specified (see 6.2.1), may be necessary to meet the properties specified.

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6. NOTES

6.1 Intended use. These castings have high abrasion resistance, excellent corrosion resistance and high hardness. They are used for bearings, bushings and seals operating under high loads with little or no lubrication. Alloy R30001 has higher abrasion resistance than alloy R30006, but with lower impact strength.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Composition required (see 1.2).
(c) If first article inspection is required (see 3.1).
(d) Type of casting, if a specific type is required (see 3.2).
(e) First article radiographic standard or criticality level if MIL-STD-278 is invoked (see 4.3.2).
(f) Level of packing and packaging required (see 5.1).
(g) If marking is in accordance with a standard other than MIL-C-3993 (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.1.2	Certification data report	UDI-A-23264	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquistion functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

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6.3 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of two castings. The contracting officer should include specific instructions in aquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Supersession data. This specification supersedes MIL-C-15345G dated 9 August 1965 as follows:

MIL-C-15345[1] Alloy number	QQ-C-390 Alloy number
1	C83600
2	C85500
3	C85700
4	C86500
5	C86100
6	C86300
8	C90300
9	C92200
10	C92300
11	C93400
12	C93200
13	C95400
14	C95500
15	C95400 or C95500
23	C91600

MIL-C-15345[1] Alloy number	QQ-N-288 Composition number
16	B
17	C
18	A
19	E
26	C

MIL-C-15345[1] Alloy number	MIL-C-20159 Alloy number
24	C96400
25	C96200

MIL-C-15345[1] Alloy number	MIL-B-24480 Alloy number
28	C95800

[1] Alloys 7 and 27 are no longer in use.

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6.5 Subject term (key word) listing.

Alloy
Bearing
Bushing
Casting
Chromium
Cobalt
Master melt
Seals
Tungsten

Custodians:
Army - MR
Navy - SH
Air Force - 20

Review activities:
Army - MI, AR, ME
Navy - OS, AS
DSC - IS

Preparing activity:
Navy - SH
(Project MECA-0393)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-C-24689A	COBALT ALLOY CASTINGS, WEAR AND CORROSION RESISTANT

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

8. DATE OF SUBMISSION *(YYMMDD)*

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